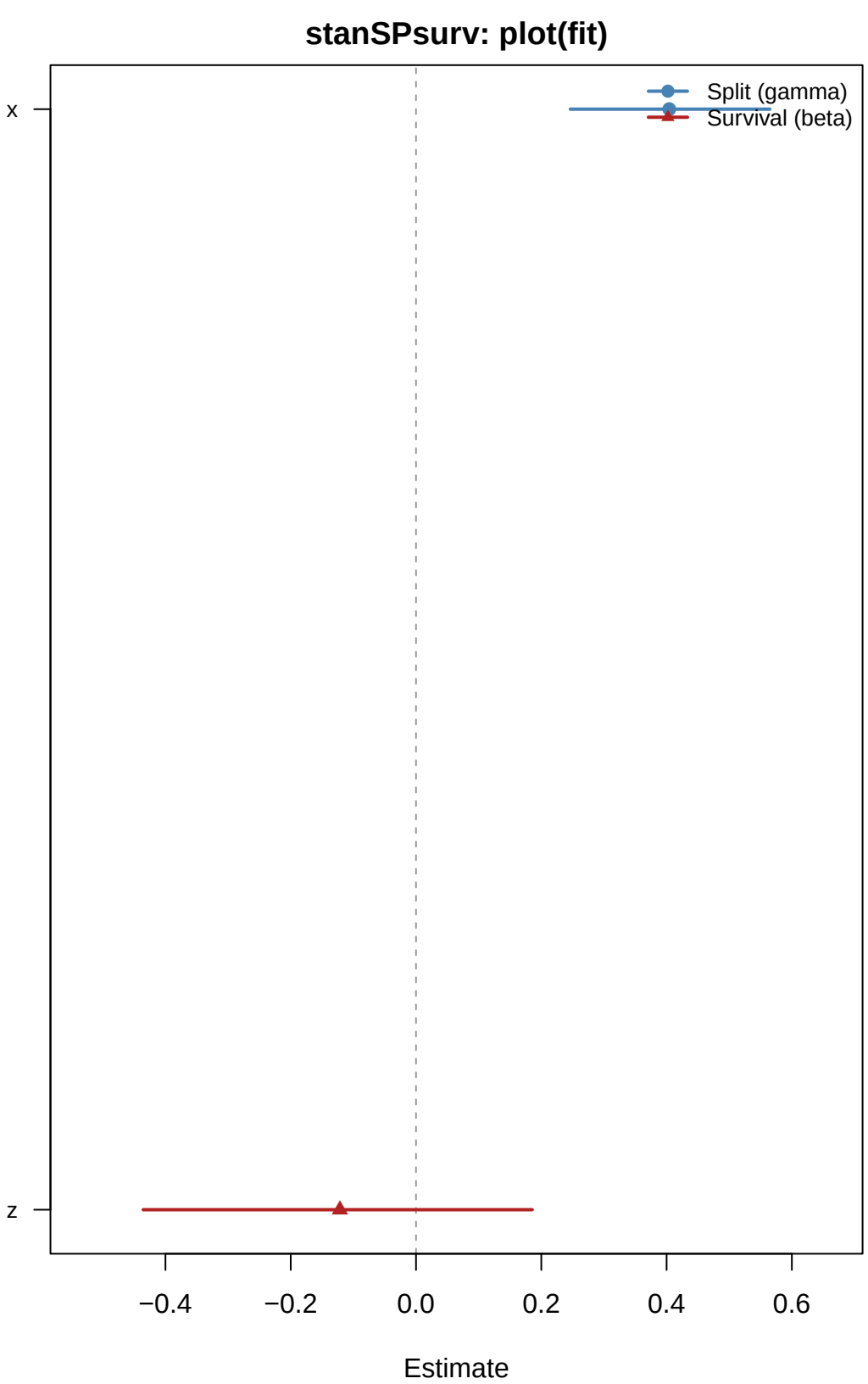
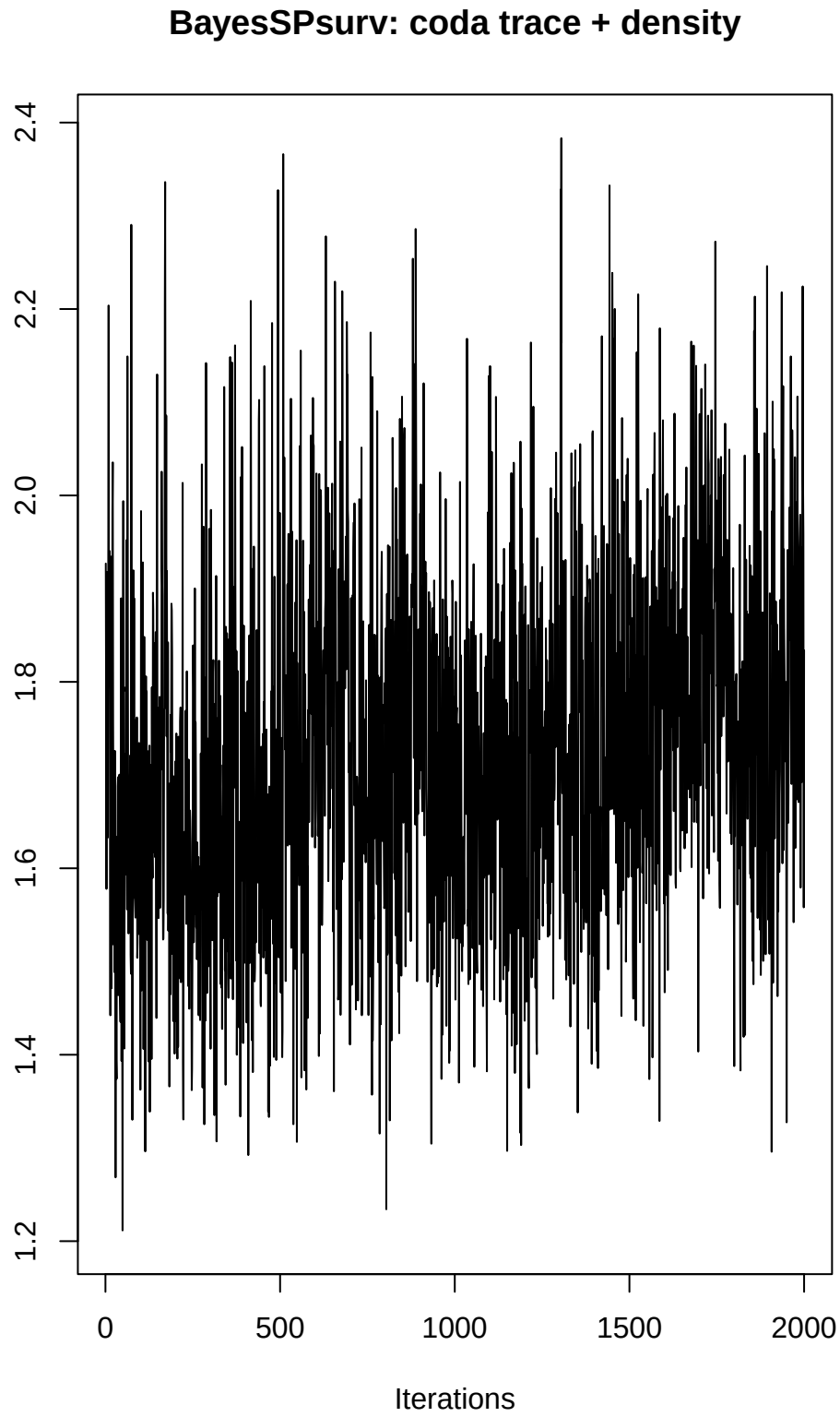
# stanSPsurv vs BayesSPsurv

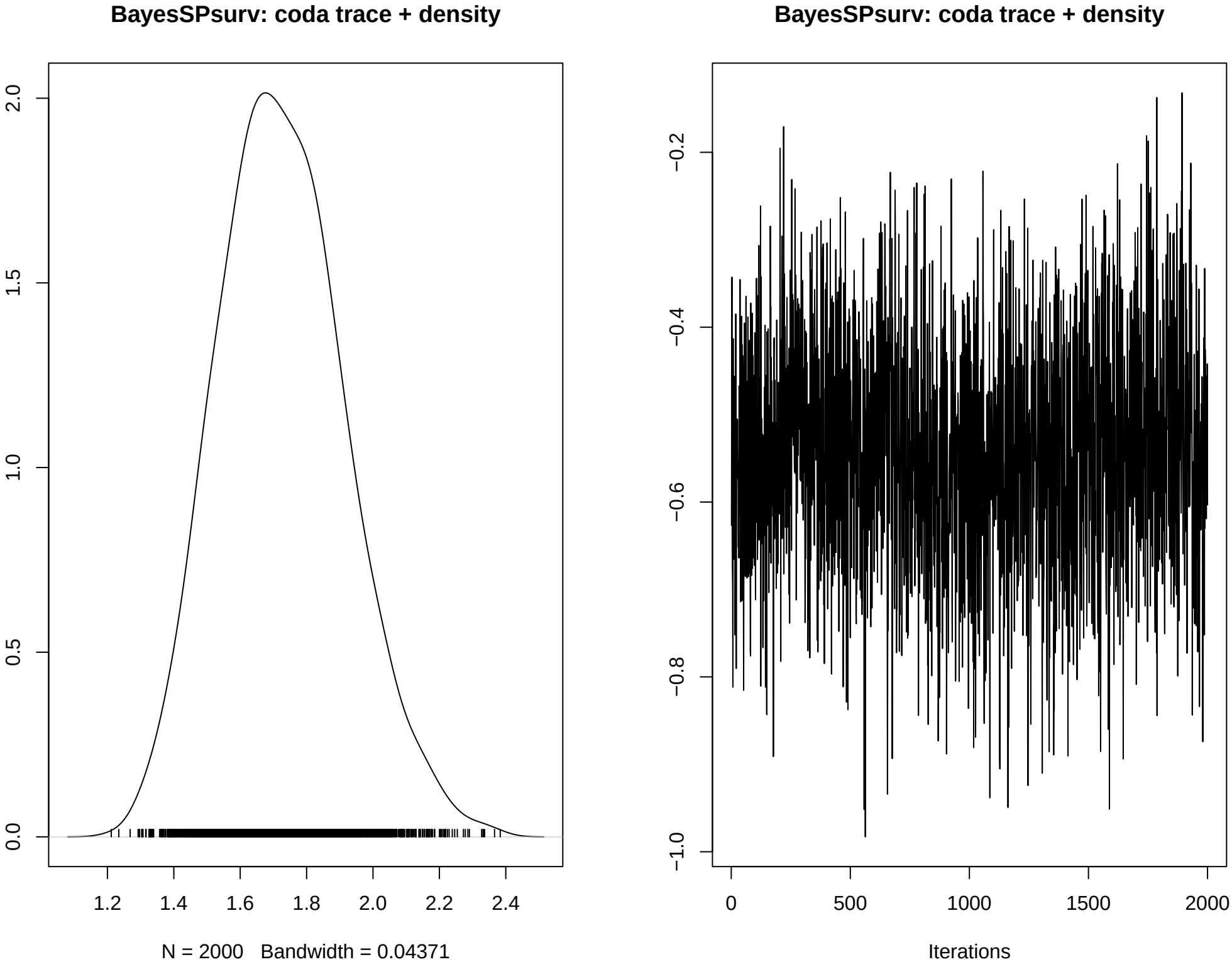
Visual comparison of plotting capabilities

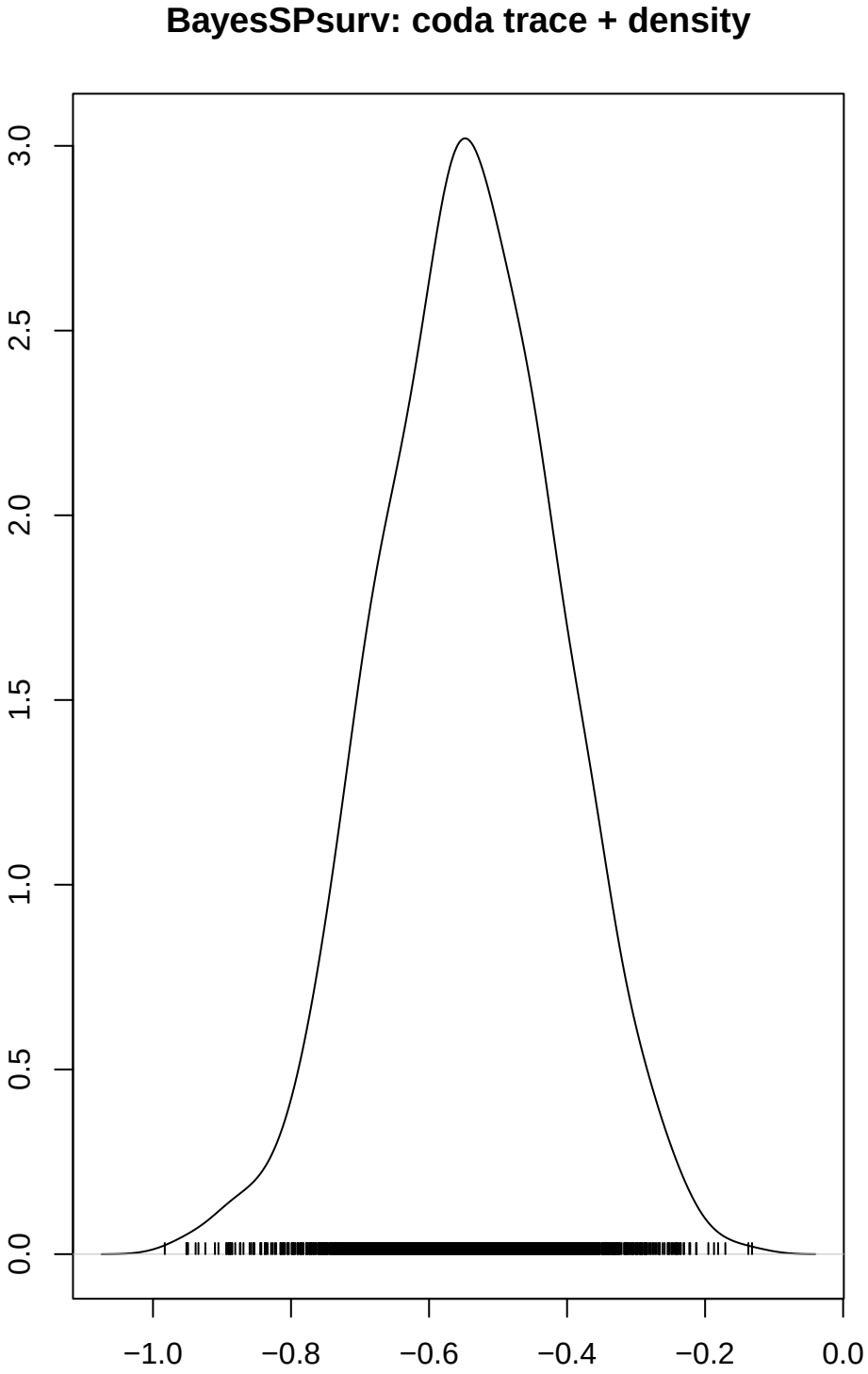
Exchangeable Weibull model, N = 400 , J = 8

True: beta = (−1.1, 0.45), gamma = (0.7, −0.5), shape = 0.75

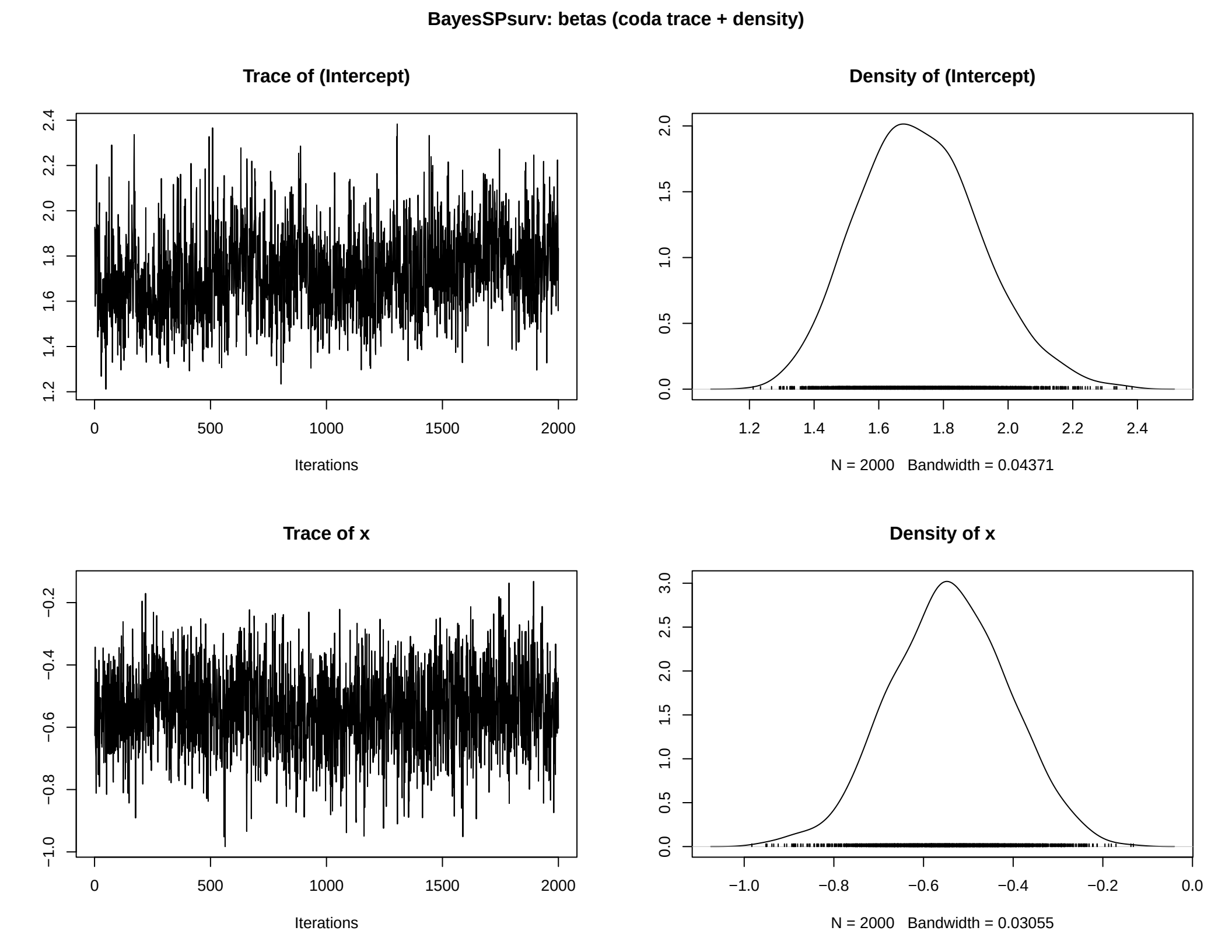

stanSPsurv: plot(fit)
Split (gamma)
Survival (beta)
x
z
−0.4
−0.2
0.0
0.2
0.4
0.6
Estimate

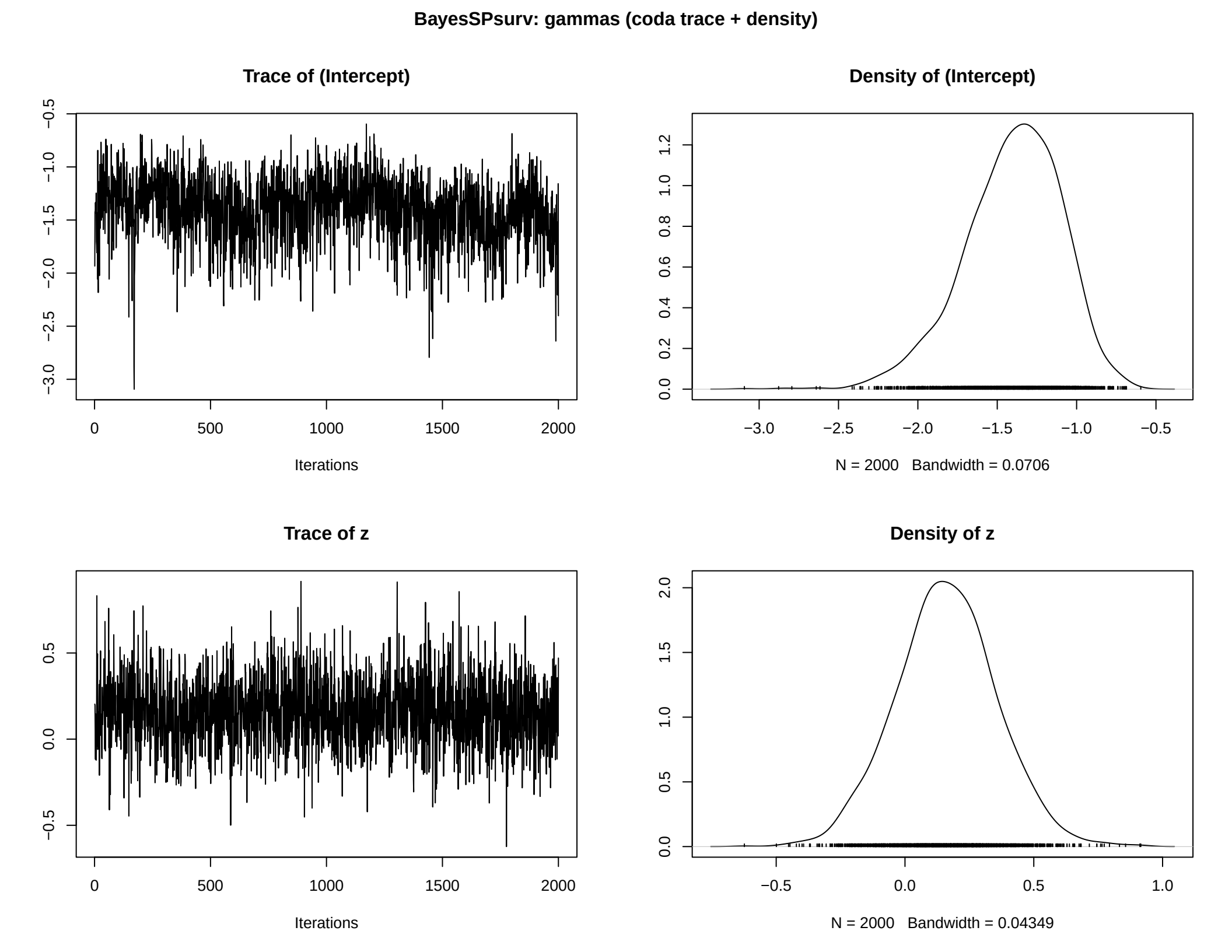

BayesSPsurv: coda trace + density
2.4
2.2
2.0
1.8
1.6
1.4
1.2
0
500
1000
1500
2000
Iterations

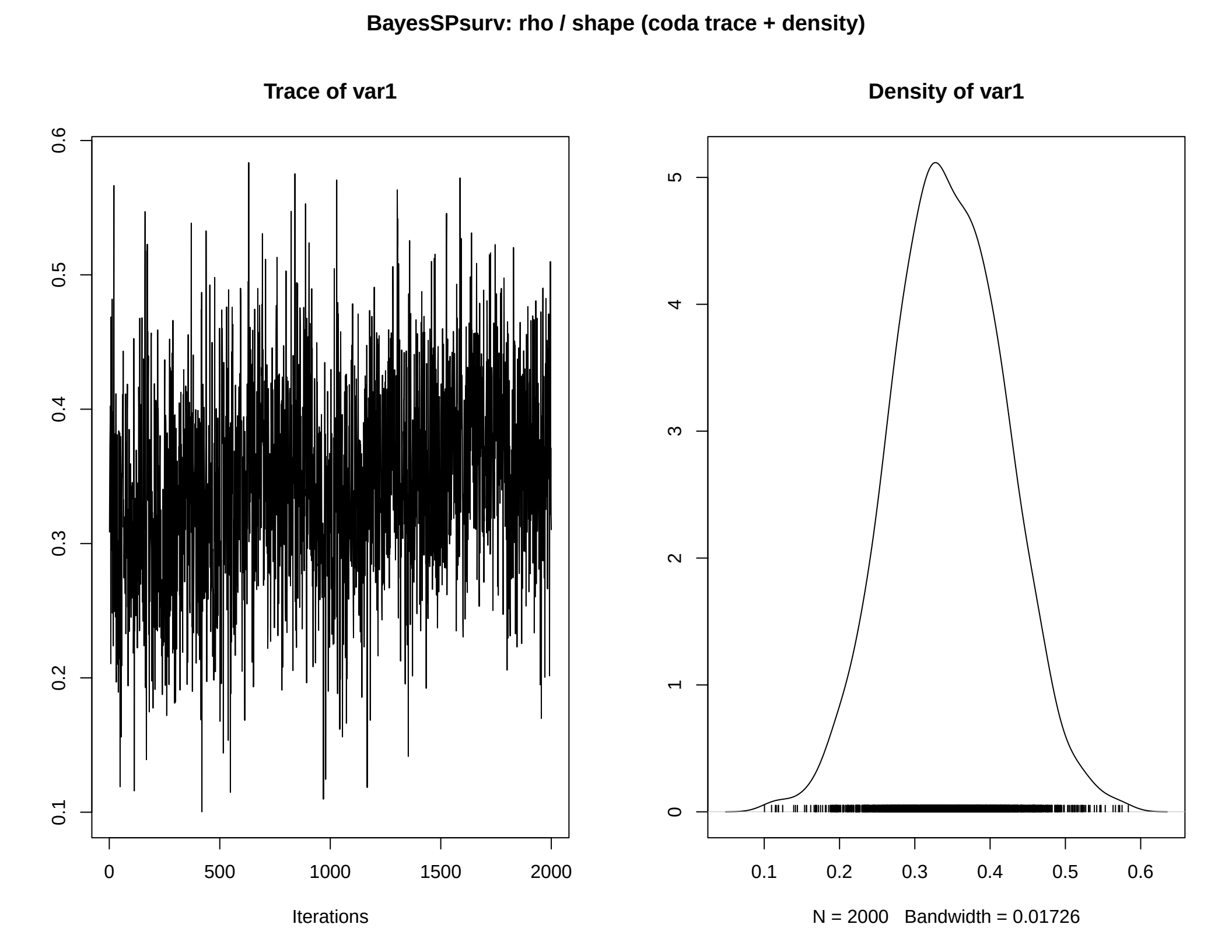

BayesSPsurv: coda trace + density
0.0
0.5
1.0
1.5
2.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
N = 2000 Bandwidth = 0.04371
BayesSPsurv: coda trace + density
−1.0
−0.8
−0.6
−0.4
−0.2
0
500
1000
1500
2000
Iterations

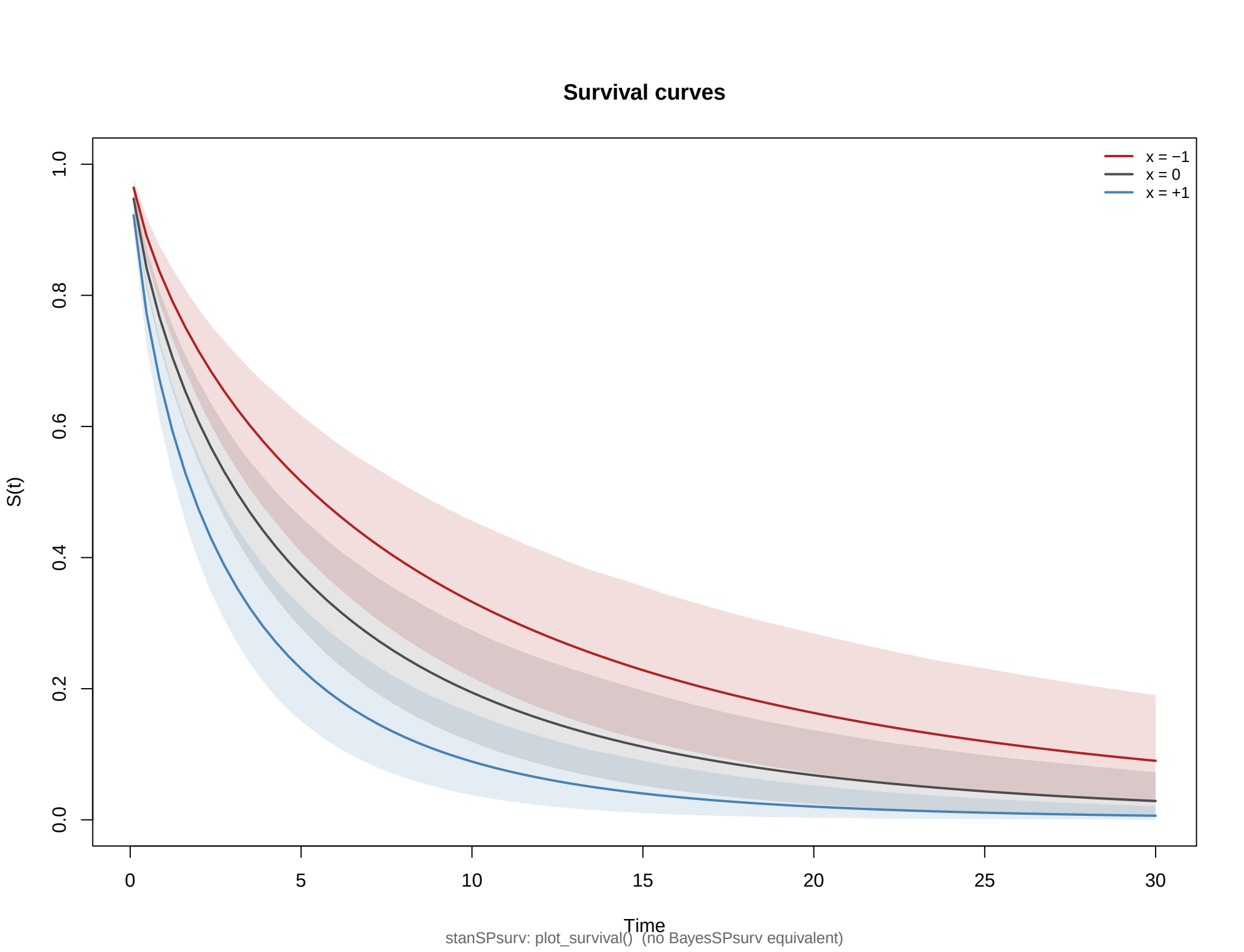


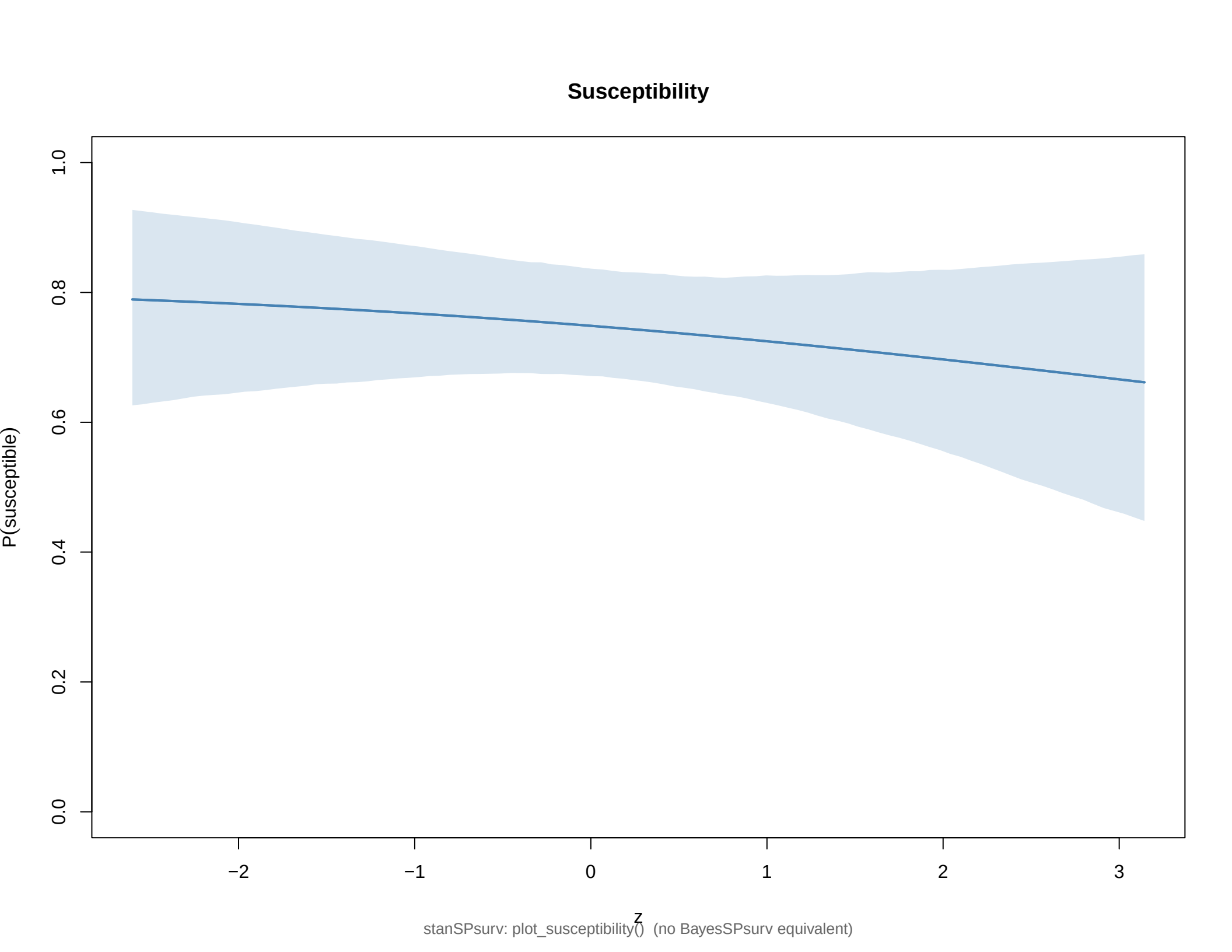

(BayesSPsurv has no coefficient plot − only coda trace/density)

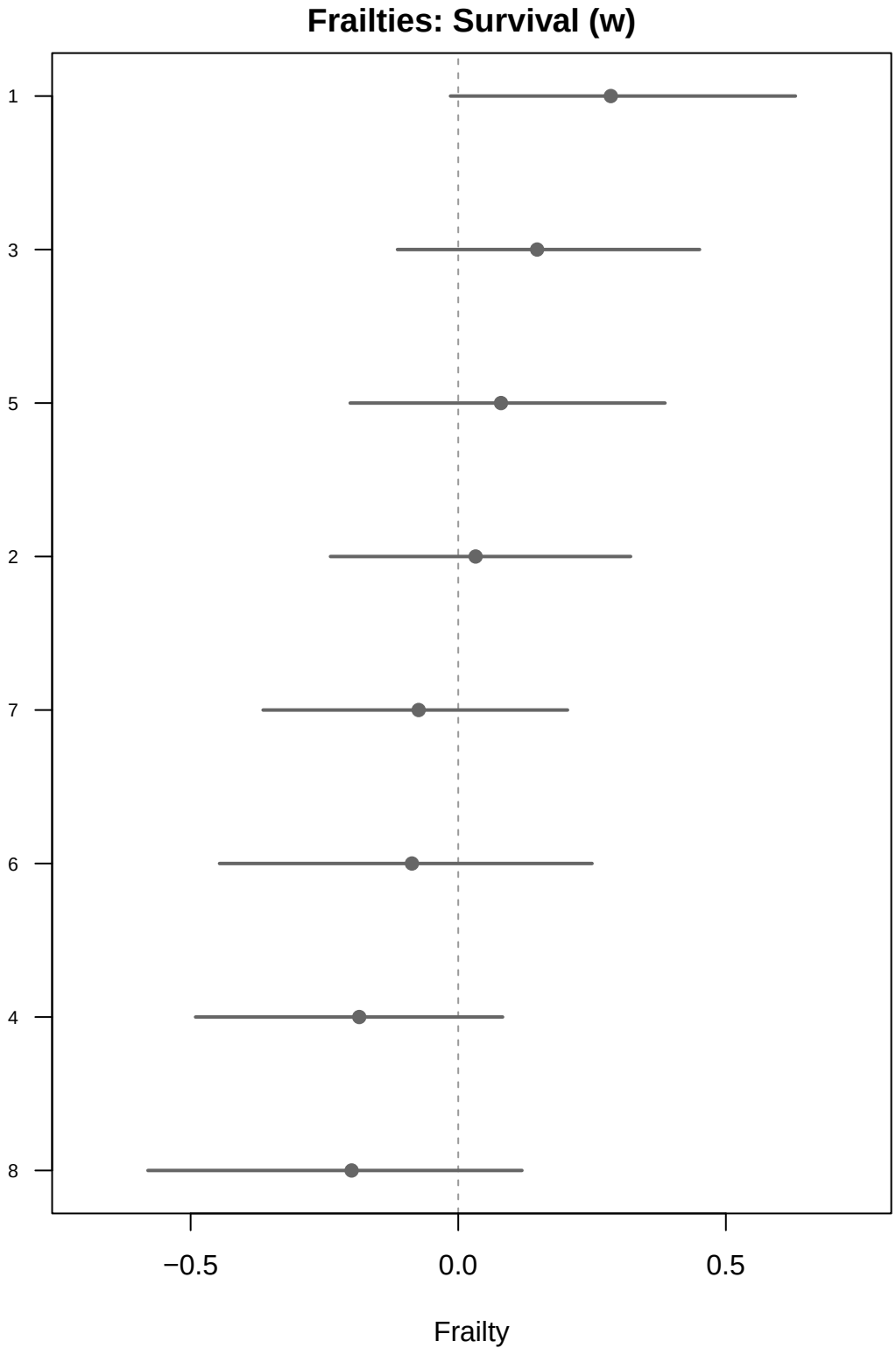

BayesSPsurv: betas (coda trace + density)
Trace of (Intercept)
Density of (Intercept)
Iterations
N = 2000 Bandwidth = 0.04371
Trace of x
Density of x
Iterations
N = 2000 Bandwidth = 0.03055

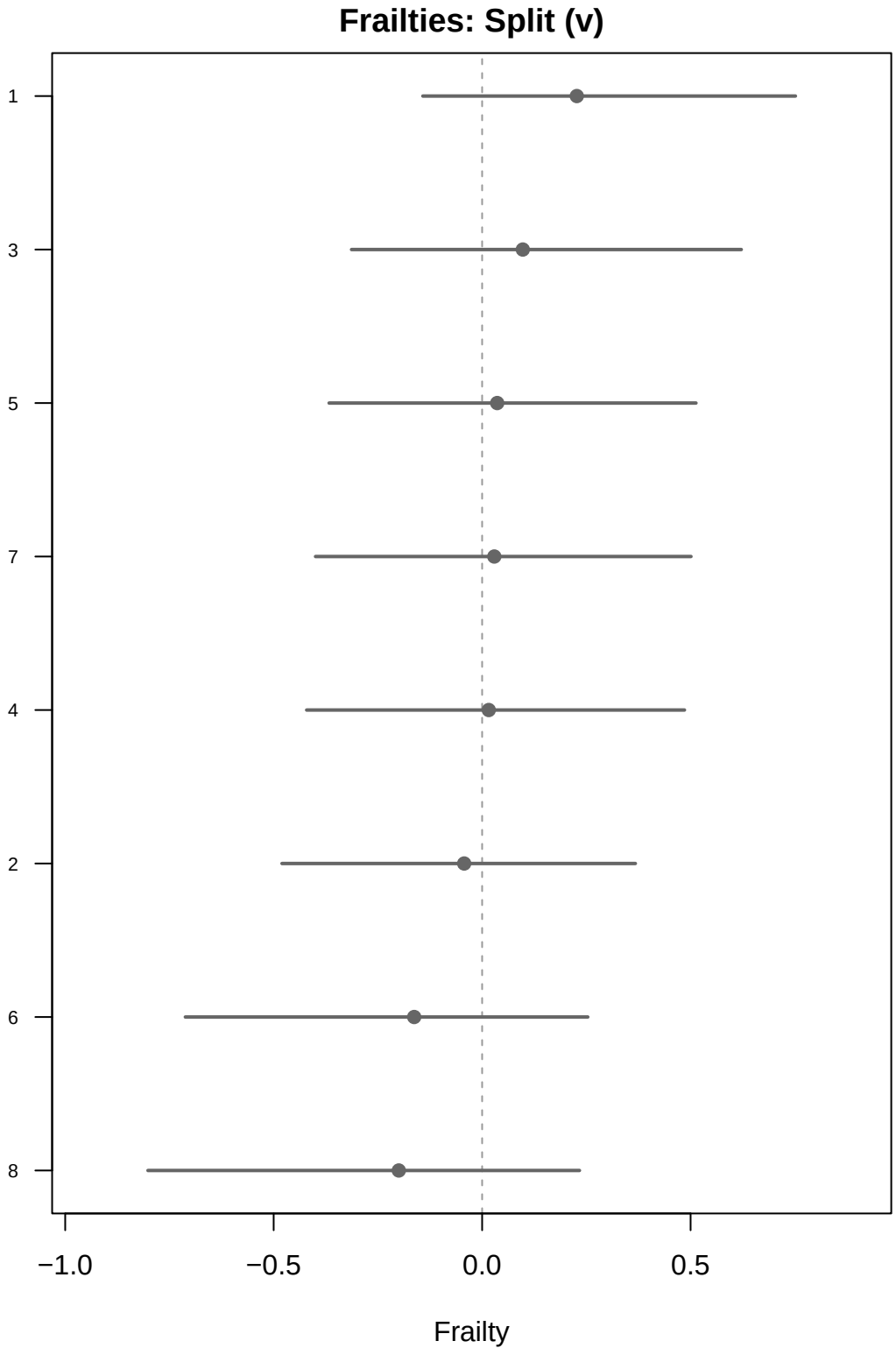

BayesSPsurv: gammas (coda trace + density)
Trace of (Intercept)
Density of (Intercept)
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
0
500
1000
1500
2000
Iterations
0.0
0.2
0.4
0.6
0.8
1.0
1.2
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
N = 2000 Bandwidth = 0.0706
Trace of z
Density of z
0.5
0.0
−0.5
0
500
1000
1500
2000
Iterations
0.0
0.5
1.0
1.5
2.0
−0.5
0.0
0.5
1.0
N = 2000 Bandwidth = 0.04349

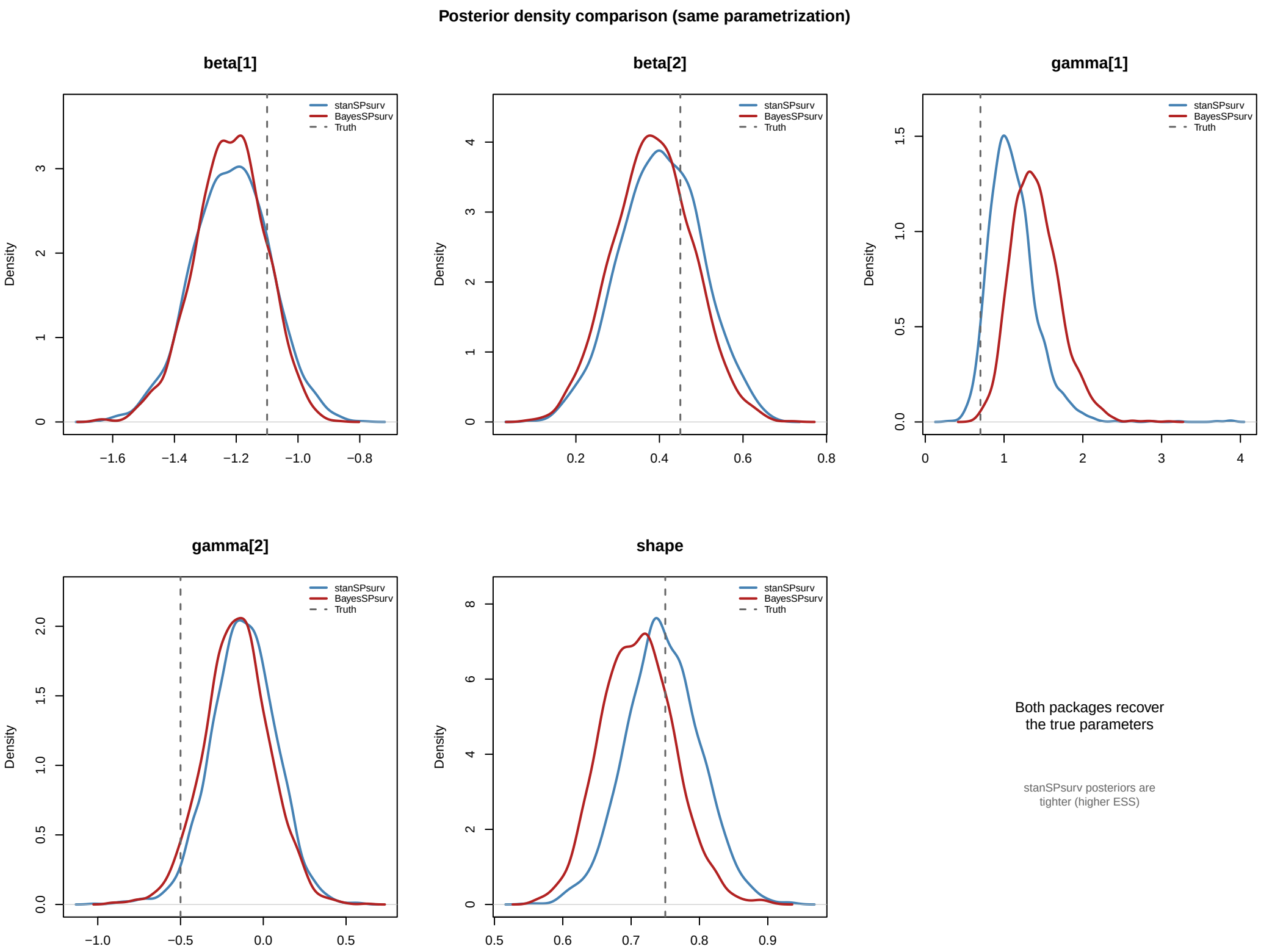

BayesSPsurv: rho / shape (coda trace + density)
Trace of var1
Density of var1
0.1
0.2
0.3
0.4
0.5
0.6
0
500
1000
1500
2000
Iterations
0
1
2
3
4
5
0.1
0.2
0.3
0.4
0.5
0.6
N = 2000 Bandwidth = 0.01726


Survival curves
x = −1
x = 0
x = +1
S(t)
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
Time
stanSPsurv: plot_survival()  (no BayesSPsurv equivalent)


Susceptibility
P(susceptible)
1.0
0.8
0.6
0.4
0.2
0.0
−2
−1
0
1
2
3
z
stanSPsurv: plot_susceptibility()  (no BayesSPsurv equivalent)


Frailties: Survival (w)
1
3
5
2
7
6
4
8
−0.5
0.0
0.5
Frailty


Frailties: Split (v)
1
3
5
7
4
2
6
8
−1.0
−0.5
0.0
0.5
Frailty


Posterior density comparison (same parametrization)
beta[1]
beta[2]
gamma[1]
gamma[2]
shape
stanSPsurv
BayesSPsurv
Truth
Density
Both packages recover
the true parameters
stanSPsurv posteriors are
tighter (higher ESS)

# Feature comparison

| Feature | stanSPsurv | BayesSPsurv |
|---|---|---|
| Coefficient forest plot | … | … |
| Survival curves with credible bands | … | … |
| Susceptibility curves | … | … |
| Frailty caterpillar plot | … | … |
| Trace plots (via coda/cmdstanr) | … | … |
| Posterior density comparison | … | … |
| Spatial autocorrelation (Moran I) | … | … |
| Join count test | … | … |
| Posterior predictive checks | … | … |
| LOO−CV / WAIC (via log_lik) | … | … |